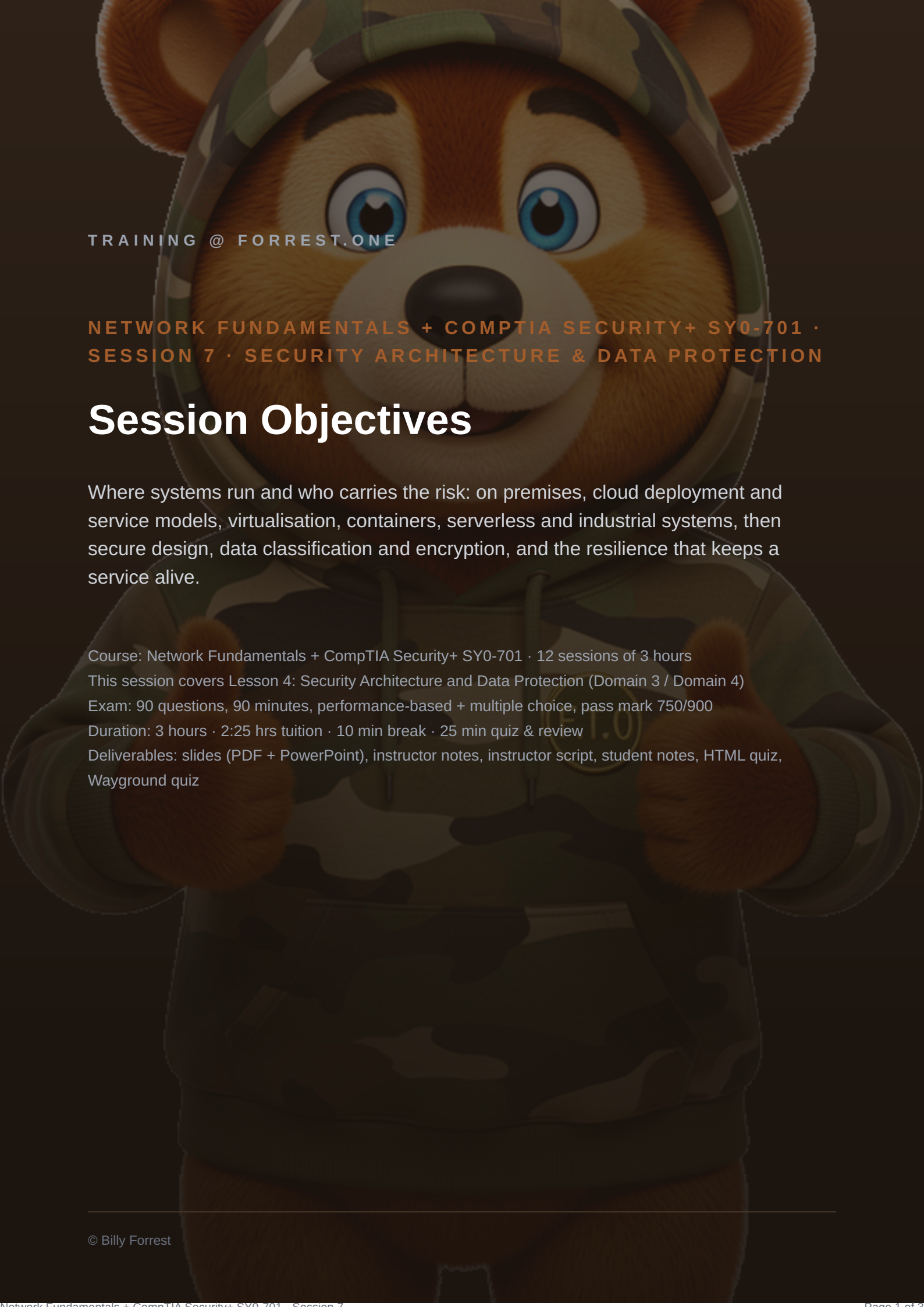




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 7 · SECURITY ARCHITECTURE & DATA PROTECTION

Session Objectives

Where systems run and who carries the risk: on premises, cloud deployment and service models, virtualisation, containers, serverless and industrial systems, then secure design, data classification and encryption, and the resilience that keeps a service alive.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session covers Lesson 4: Security Architecture and Data Protection (Domain 3 / Domain 4)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives — Security Architecture & Data Protection

This session covers the architecture that everything else stands on: where systems run, who carries the risk in each model, how platforms are designed and contained, and how data is identified, classified and protected in every state it occupies. It closes with the resilience and continuity planning that keep a service alive when hardware fails.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Describe on-premises architecture, including physical control, responsibility and total cost of ownership.
- Compare cloud deployment models: public, private, hybrid and community, and the security responsibility in each.
- Explain cloud service models: infrastructure, platform and software as a service, and how the responsibility line moves.
- Describe virtualisation security: hypervisor types, virtual machine escape, sprawl and snapshots.
- Explain containerisation and serverless architecture, including image risk and the event trigger as an entry point.
- Describe internet of things risk, and industrial control systems with SCADA and the IT and OT convergence problem.
- Explain infrastructure as code: its benefits, and misconfiguration at scale and secrets in files as its risks.
- Describe secure network design: defence in depth, segmentation, access control lists, the demilitarised zone and least connectivity.
- Justify control selection against risk, compliance, cost and operational impact.
- Identify data types that carry legal weight, and the three states of data.
- Explain classification levels and how classification drives storage, transmission and disposal.
- Describe protection of data at rest and data in transit, including key management and hardware security modules.
- Explain high availability, alternate sites, backup strategies and the recovery point and recovery time objectives.
- Explain power and environmental controls, platform diversity, resilience testing, and business continuity planning.

CompTIA exam context

Item	Detail
Exam	Network Fundamentals + CompTIA Security+ SY0-701
Objective area	Architecture and data protection objectives across Domain 3 (Security Architecture) and Domain 4 (Security Operations)
Weight	Architecture concepts appear across multiple domains, and are consistently tested
Pass mark	750 on a scale of 100 to 900
Style	Scenario questions: where does the responsibility line sit, which control fits this data state, what recovery objective applies

Lesson 4 coverage

Area	What is covered
Architecture models	On premises; public, private, hybrid and community cloud; IaaS, PaaS and SaaS
Compute and edge	Virtualisation and the hypervisor; containers; serverless; internet of things; ICS and SCADA; IT and OT convergence
Secure design	Infrastructure as code and its risks; secure network design; protecting the channel; control selection
Data protection	Data types and states; classification levels and handling; data at rest; secure disposal; data in transit
Resilience and recovery	High availability; alternate sites; backup strategies and recovery objectives; power and environment; testing; business continuity

How this session maps to the exam

Architecture questions reward a single habit: work out who owns the risk, then work out what the data needs. Use the session quiz to 70% or better before the exam.